

An executable reduction from planar 3-SAT to geometric connected domination

Reconstruction of published constructions · Awaiting expert review
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Abstract

We reconstruct a deterministic polynomial-time reduction from embedded planar 3-SAT to connected domination on explicitly given integer points. The reduction composes variable cycles and clause triangles, a planar connected-cover augmentation, and the grid construction of Clark, Colbourn and Johnson. A visibility representation with a finite catalog of routing templates supplies explicit coordinates. Recovery normalizes an arbitrary valid target witness, then extracts a satisfying assignment; it also handles the no-solution output. The construction uses quadratically many points with logarithmic coordinate bit length. The contribution is an executable, inspectable reconstruction with independent finite checks for a known hardness result.

1. Introduction

A connected dominating set selects vertices that both cover a network and communicate through selected vertices. Geometric graphs make the representation part of the problem: coordinates must create the required edges without adding unintended ones. A hardness statement alone does not specify a usable map from formula encodings to point sets or a decoder for arbitrary target witnesses.

Lichtenstein’s planar constructions [L82] and the connected-cover augmentation of Garey and Johnson [GJ77] provide combinatorial stages. Clark, Colbourn and Johnson [CCJ90, Section 6] prove connected-domination hardness for grid graphs. Tamassia and Tollis [TT86] supply a visibility representation from which an orthogonal drawing can be constructed. We combine these ingredients into one reproducible rule, including deterministic tie-breaking, degenerate inputs, integer coordinates, and recovery of every valid output. We do not claim that this is the first implementation of these constructions.

Theorem 1 (Executable reconstruction). Let φ be a three-literal CNF formula with a planar rotation system of its variable–clause incidence graph. There are deterministic polynomial-time maps F and G such that $F(\varphi)$ is a legal rational-point connected-domination instance and

$$y \in S_B(F(\varphi)) \implies G(\varphi, y) \in S_A(\varphi). \quad (1)$$

Here S_A and S_B are the valid-output sets defined below, including the no-solution output. If φ has n variables and m clauses, $F(\varphi)$ uses $O((n + m)^2)$ integer points, with $O(\log(2 + n + m))$ bits per coordinate.

The proof follows the three graph interfaces in Section 3, Section 4 and Section 6. Section 5 gives the coordinate construction. The supplied implementation is the algorithm of the theorem; the verification appendix records its checked revision and evidence limits.

2. Instances and output semantics

A source instance specifies n variables, m clauses of exactly three signed literal occurrences, and a clockwise rotation at every incidence vertex. Repeated literals remain separate occurrences but induce one incidence edge. Empty formulas and isolated variables are allowed. A valid source output is a Boolean vector of length n satisfying every clause. The distinguished answer NO-SOLUTION is valid exactly when no such vector exists. All graphs and rotations are explicit, so their encoding includes the isolated variables.

A target instance is a nonempty finite set $P \subset \mathbb{Q}^2$ of distinct points and a nonnegative integer K . Two points are adjacent exactly when their squared Euclidean distance is at most one. A valid witness is

a subset $D \subseteq P$ with at most K points, such that every point lies in D or has a neighbor in D , and the graph induced by D is connected. The empty set is not connected here. The answer NO-SOLUTION is valid exactly when no witness exists. A target witness is encoded by distinct indices in the ordered point list.

The forward map first runs ordinary unit propagation, discarding repeated literals within a clause for that check and ignoring tautologies. A derived contradiction yields the one-point instance with $K = 0$. Otherwise the formula used below is unchanged. An empty formula with no variables yields one point with $K = 1$. These branches are polynomial and preserve the output semantics.

3. From planar SAT to planar cover

For a variable with d occurrences, make an even cycle with $2 \max(2, d)$ vertices. Its alternating classes represent true and false. Allocate a distinct pair of cycle positions to each occurrence in the supplied incidence order, and attach the occurrence to the vertex of its literal's parity. For each clause, make a triangle with one vertex per occurrence and join that vertex to its literal port. Parallel occurrences along one incidence edge are expanded in opposite orders at its ends. This preserves the planar rotation. Denote the resulting graph by Q and its edge count by e_Q . Every vertex of Q has degree two or three. Set

$$B = \sum_{i=1}^n \max(2, d_i) + 2m. \quad (2)$$

Figure 1 shows the complete gadget for one repeated-literal clause. The use of distinct ports is relevant even when the incidence graph has only one edge between the variable and the clause.

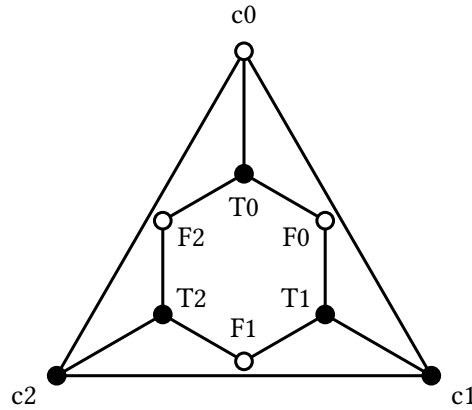


Figure 1: A six-cycle for one variable and a triangle for the clause $(x \vee x \vee x)$. Each T_i has its own clause edge; F_i is the opposite parity. Filled vertices choose the true parity and two triangle vertices, forming a cover of size five. All nine vertices and twelve edges are shown.

Lemma 2 (Lichtenstein's cover interface). The formula is satisfiable if and only if Q has a vertex cover of size at most B . Every such cover determines a satisfying assignment.

Proof. A satisfying assignment selects one parity of each variable cycle. In each clause triangle, omit a vertex whose literal is true and select the other two. Every cycle and triangle edge is covered, and the edge incident with the omitted triangle vertex is covered by its selected literal port. The count is Equation 2.

Conversely, a cycle of length $2r$ needs at least r vertices in any cover, and each triangle needs two. The disjoint cycle and triangle subgraphs already require B vertices. A cover within the bound therefore meets every local lower bound exactly. A cover of an even cycle at this lower bound is one of its two alternating classes: a missing vertex forces both neighbors, and propagating the equality in the bound forces alternation around the cycle. Read that parity as the variable value. Each omitted triangle vertex

forces its literal port into the cover, so each clause is satisfied. The four-cycle for an isolated variable gives either legal Boolean value. This is the explicit interface used from [L82, Section 4, Theorem 3]. $\square$

4. From planar cover to connected cover

Replace each edge uv of Q by the path u, a, b, v . Enumerate the faces of a planar embedding. For each face, and for each distinct boundary vertex whose current degree is below four, add a spoke to a new vertex w and add a pendant neighbor of w . Join the new w vertices of that face in their cyclic boundary order. For disconnected Q , merge one chosen outer face per component into a single exterior boundary before adding its cycle. All enumeration follows fixed insertion order. Write H for the augmented graph and r for the number of spoke vertices w . Its connected-cover budget is

$$B' = B + e_Q + r. \quad (3)$$

The augmentation is illustrated in Figure 2 on a triangle. In the actual reduction, the input is the cycle-and-triangle graph Q above. Each face cycle has at least three distinct spokes; subdivision provides boundary vertices even when an original edge borders the same face twice.

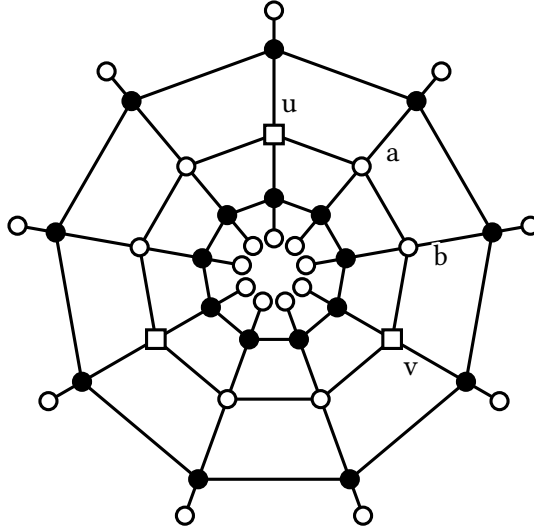


Figure 2: Complete face augmentation of a triangle after two subdivisions per edge. Squares are original vertices; the other unfilled vertices on the middle ring are subdividers. Filled vertices are the two face cycles, each with spokes and pendant neighbors. The labeled path is u, a, b, v . All 45 vertices and 63 edges are shown. This example illustrates both an interior face and the exterior face.

Lemma 3 (Connected-cover interface). The graph H is connected, planar, and has maximum degree four. A cover of Q of size at most B extends to a connected cover of H of size at most B' . Every connected cover of H within that bound yields a cover of Q within B .

Proof. Face cycles and their spokes are drawn within their faces; the common exterior cycle joins distinct components. Every spoke consumes an available incidence at its boundary vertex. A new w has two cycle neighbors, one spoke neighbor and one pendant neighbor, giving degree four.

Given a cover of Q , choose a on a subdivided edge when u is selected and choose b otherwise; in the latter case v is selected. Add every w . This covers each subdivided edge and all new edges. The face cycles connect through selected subdividers across shared edges, and the selected original vertices have spokes to these cycles. The merged exterior cycle handles disconnected components. This is the connected extension of [GJ77, Lemma 2], with one subdivider per original edge and all r spoke vertices.

In the reverse direction, a connected cover must contain each w . Omitting it would force its pendant neighbor into the cover and isolate that selected neighbor from the other selected vertices. Every subdivided edge uses at least one of a, b . If neither original endpoint is selected, it uses both. Keep the

selected original vertices and add one endpoint of each such uncovered original edge. Charging each added endpoint to the extra subdivider gives at most $B' - r - e_Q = B$ vertices. Extra selected pendants can only reduce this available budget. $\square$

The leaf-free core of H is biconnected. To see the needed premise, delete one original vertex or subdivider. Its neighbors reconnect around the face cycles between consecutive incident edges. When an adjacent original vertex lacks a spoke on that face, the next subdivider on its other incident edge supplies one. Thus the subdivided components remain joined, with the exterior cycle also joining distinct original components. If a face-cycle vertex is deleted, the remainder of its cycle stays connected, and its spoke neighbor reaches that remainder through another boundary subdivider. These exhaust the core's vertex types. The core has at least three vertices.

5. Explicit orthogonal coordinates

We describe the total coordinate construction before its bounded compact implementation option. Let v_H be the number of vertices in the core. Choose its first edge st . Biconnectivity supplies an s - t path after removing this edge. Start an order with that path. While vertices remain, choose the first outside component and its first two neighbors a, b in the current order. Biconnectivity ensures that these neighbors are distinct. Insert the internal vertices of an a - b path through the component directly after a . Materialize induced graphs in parent node and edge order, so all component and shortest-path choices have fixed tie-breaking.

Each inserted vertex has an earlier and a later neighbor; old vertices retain those neighbors. The final order orients the edges acyclically with unique source s and sink t . Use the face to the left of st as the exterior. The plane st -graph visibility theorem [TT86, Algorithm W-VISIBILITY, p. 328; Theorem 1, p. 330] gives consecutive incoming and outgoing incidences and an acyclic left-to-right face dual when the arc for st is omitted. Let ρ be distinct topological ranks of that dual and let η be the vertex order. Draw each non- st edge vertically at the rank of its left face, between its endpoint heights η . Draw st at column -1 . Each vertex bar spans its incident columns.

These coordinates instantiate the cited visibility construction. In its strict-potential formulation, take face potentials 2ρ and edge column $2\rho(\text{left}) + 1$, with exterior edge column -1 , then apply the affine map $x \mapsto \frac{x-1}{2}$. Every dual inequality remains strict. Tightening a bar to the span of its incidences creates no intersections, including when that span is a single point. Coordinates and the number of bars are $O(v_H)$.

To replace bars by point vertices, list their distinct columns from left to right and record whether each has an incidence above, below, or both. With degree two through four there are 46 patterns. A fixed certificate for each pattern consists of simple orthogonal paths from one center to terminals at $(2i + 1, \pm 4)$ for $i = 0, \dots, k - 1$, where k is the number of columns. The paths lie in $[0, 2k] \times [-4, 4]$. Paths intersect only at the center and meet the strip boundary only at their terminals. The complete finite catalog is supplied with an exhaustive independent validator. It is data used by the algorithm; no solver runs to choose a template.

Multiply bar coordinates by 20. Map odd template abscissas to the actual incidence columns, internal even abscissas to the midpoints of consecutive columns, and exterior abscissas to four units beyond the extreme columns. This strictly increasing integer map preserves disjointness and orthogonality. Translate heights by the scaled bar height and join matching terminals along the original vertical segments. Vertex strips are at least twelve units apart. A nonincident edge is at least twenty units beyond the unextended bar span, so the four-unit margin cannot reach it. The resulting drawing has $O(v_H)$ bends, $O(v_H)$ coordinate magnitude and $O(v_H^2)$ total lattice route length.

The executable rule first permits a compact TSM drawing only for cores with at most 1,000 vertices and at most 50,000,000 Python line events. Vertices are renumbered by insertion order before this call, preventing hash-dependent set choices in the locked planar-drawing dependency. An exception or exhausted allowance selects the visibility construction above. A compact result is accepted only after checking integer coordinates of magnitude at most $100v_H$, at most $100v_H$ drawn vertices, and

exact topology and nonintersection. The finite operation allowance is independent of elapsed time. This bounded option affects coordinates but leaves the proved drawing interface unchanged.

6. From connected cover to geometric domination

Scale the core drawing by ten. Restore each original pendant vertex four grid steps from its neighbor in the first unused compass direction. Insert every integer point on each routed edge. Let V_1 be the points for the vertices of H , and let V_2 be the interior path points. On each path its first and last interior points are *connectors*; the other interior points are *middle points*. Every path has at least three interior points.

For each middle point, in lexicographic order, add the first grid neighbor in east, north, west, south order that has that middle point as its unique main-path neighbor. Call the new set V_3 . Scale-ten separation makes these choices available and distinct. A side point has no original-vertex or connector neighbor. Its other side neighbors, at most two, belong to adjacent middle points on the same path. Figure 3 displays the straight-path case. At bends the same local adjacency conditions hold; separated strips and routed edges prevent contacts with unrelated paths. The implementation checks all these adjacencies exactly.

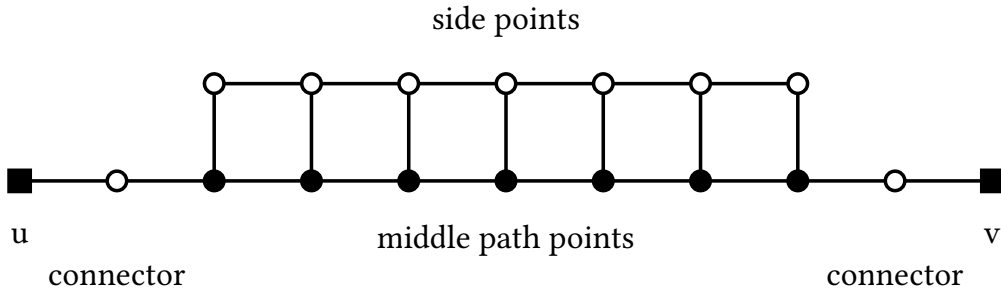


Figure 3: The exact grid graph of one straight route of length ten. Squares are original endpoints, unfilled lower circles are connectors, filled lower circles are middle points, and the upper row contains their side points. Every unit-distance edge is drawn. Side points may form a chain; they are not assumed to be isolated leaves. Other routes incident with u and v are omitted from this local illustration.

Write $N_1 = |V_1|$, $N_2 = |V_2|$, and $e_H = |E(H)|$. Emit

$$P = V_1 \cup V_2 \cup V_3, \quad K = B' + N_2 - e_H + N_1 - 1. \quad (4)$$

The points are distinct integers. On integer points Euclidean distance at most one is exactly grid adjacency, so the described graph is the target graph. The budget is nonnegative: every path contributes at least three points to V_2 and H is nonempty.

Lemma 4 (Forward domination witness). A connected cover C of H gives a connected dominating set of size $|C| + N_2 - e_H + N_1 - 1$.

Proof. Choose a spanning tree of $H[C]$ and attach each omitted vertex to a neighbor in C . This gives a spanning tree on all N_1 vertices. Select the points for C , all middle points, both connectors of each tree edge, and one connector incident with C on every other edge. The selected points are connected along this tree. They dominate omitted originals, omitted connectors and all side points. There are $N_2 - 2e_H$ middle points and $e_H + N_1 - 1$ selected connectors, giving the count. $\square$

Lemma 5 (Recovery from arbitrary target witnesses). Every connected dominating set D of size at most K can be transformed, without increasing its size, into one containing no side points. Its selected original vertices then form a connected cover of H of size at most B' .

Proof. We use the constructive replacement argument of [CCJ90, Lemma 6.1]. Pair each middle point with its side point. If every pair on a route contains a selected point, replace selected side points by the corresponding middle points. If a pair is missing, its side point must be dominated by a selected neighboring side point. A selected connection to an endpoint crosses the pairs on one side of that missing pair. It uses both layers in at least one pair along the connection, which pays for the extra middle point at the missing pair. Replace the selected side points on that prefix or suffix by middle points and add

the missing middle point. The replacement preserves connectivity and domination without increasing cardinality.

The decoder enumerates these prefixes and suffixes in route order and checks cardinality, domination and connectivity before accepting one. The preceding argument supplies an acceptable move whenever a side point remains. Each move strictly decreases the number of selected side points, so the process ends. This makes the minimal-set argument in the cited lemma into an explicit algorithm and applies to every valid D , not only an optimum witness.

Without selected side points, every middle point is selected, since it must dominate its side point. Each route needs a connector incident with a selected original endpoint. Thus the selected originals form a cover C of H . Their connectivity can only pass between originals through complete routes, so $H[C]$ is connected, as in [CCJ90, Lemma 6.2]. Every omitted original needs an additional selected connector on an incident edge to be dominated, and connecting C needs at least $|C| - 1$ edges with both connectors selected. These edge classes are disjoint. Consequently the normalized witness satisfies

$$\begin{aligned} |D| &\geq |C| + (N_2 - 2e_H) + e_H + (N_1 - |C|) + (|C| - 1) \\ &= |C| + N_2 - e_H + N_1 - 1. \end{aligned} \tag{5}$$

Together with Equation 4 this gives $|C| \leq B'$. $\square$

7. The complete rule and its bounds

To recover a source witness, reconstruct the identical point list from the source, interpret the selected indices, apply Lemma 5, use the endpoint charging in Lemma 3, and read the variable parities in Lemma 2. The empty-source branch returns the empty Boolean vector. The contradictory-unit branch has no valid witness to decode. If the target answer is NO-SOLUTION, return that answer for the source: Lemmas 2–4 show that a source witness would have produced a target witness, and Lemma 5 supplies the converse implication. This proves Theorem 1 for all legal source instances and all valid target outputs.

The graph sizes through H are linear in $n + m$. The visibility algorithm uses at most v_H ear insertions and polynomial graph searches; its template catalog is constant size. Its route length is quadratic. The accepted compact option has the same asymptotic output bounds and bounded work. Scaling and adding side points preserve the quadratic size and logarithmic coordinate bit length. All comparisons are exact on these integers.

Recovery rebuilds this polynomial-size instance. It accepts at most $|V_3|$ normalization moves. Each pass enumerates polynomially many route intervals and performs explicit finite graph checks, each polynomial in $|P|$ and the coordinate bit length. The remaining cover and parity extraction is polynomial. Thus both maps have worst-case polynomial running time in their respective input encodings, with the forward encoding bound stated in Theorem 1.

Determinism includes dependency behavior. Induced visibility graphs preserve insertion order; compact inputs use small integer labels; the locked flow implementation enumerates nodes and edges into indexed arrays. Generated labels enter ordered dictionaries. Grid routes follow ordered graph traversal, and output points are lexicographically sorted. Arbitrary traversal used only to test connectedness affects a Boolean predicate, not point ordering. The same source therefore gives the same indexed coordinates in separate forward and recovery processes. Dependency upgrades require a renewed audit of this implementation correspondence.

8. Conclusion

The reconstructed rule supplies explicit integer coordinates and an executable decoder for arbitrary valid target outputs. It realizes the known grid-graph hardness route with a total polynomial drawing construction. The implementation is intended for inspection and reproduction: the quadratic expansion has large constants, and no practical large-instance solver advantage is claimed. Independent finite checks support the implementation, while the general claim rests on the composed arguments and cited interfaces. Expert mathematical review and a formal proof certificate remain distinct from these checks.

References

- [L82] D. Lichtenstein. *Planar formulae and their uses*. SIAM Journal on Computing 11(2), 329–343, 1982.
- [GJ77] M. R. Garey and D. S. Johnson. *The rectilinear Steiner tree problem is NP-complete*. SIAM Journal on Applied Mathematics 32(4), 826–834, 1977.
- [CCJ90] B. N. Clark, C. J. Colbourn and D. S. Johnson. *Unit disk graphs*. Discrete Mathematics 86(1–3), 165–177, 1990.
- [TT86] R. Tamassia and I. G. Tollis. *A unified approach to visibility representations of planar graphs*. Discrete & Computational Geometry 1, 321–341, 1986.

A. Verification and reproducibility

The checked candidate is commit 72e1365, with implementation repair at ad75146. The independent assessment at `reviews/deterministic-repair/review.md` returned advance and reused the unaffected mathematical findings from the first review. This is an agent assessment for expert review, not human certification or publication acceptance. The review began in a fresh registered reviewer context; its isolation boundary was instructional rather than a filesystem sandbox. The focused follow-up retained that review’s findings.

The original prepared corpus contains 112 explicit sources, up to six variables, including 100 seeded random cases and twelve edge cases. Source labels were obtained by exhaustive independent enumeration. The current complete forward/target-solve/recovery loop passed 195 witness outputs and fourteen exact negative outputs. Each of the 97 nontrivial geometric positive targets supplied two distinct witnesses. The empty source supplied one. Negative targets in this corpus are the actual one-point outputs of the unit-contradiction branch. The largest prepared target has 202,170 points.

Two further unsatisfiable planar sources outside the prepared corpus survive unit propagation. Their public targets have 31,576 and 25,236 points. A target-only checker verifies the complete grid-path and side-point interface from their integer coordinates. Applying the reverse implication of [CCJ90, Lemmas 6.1–6.2] reduces a hypothetical connected dominating set to a connected vertex cover with budget 308. In each recovered skeleton, articulation vertices, exact degree-two folds, and exhaustive search on the final eighteen vertices give the lower bound $240 + 57 + 12 = 309 > 308$. The two NO-SOLUTION answers therefore have structural certificates. These use the published implication; they are not unrestricted direct connected-domination solver runs.

The target-only witness finder receives coordinates, not source assignments or candidate metadata. It recognizes a subdivided backbone, uses cover folding and search, and proposes selected point indices. An independent exact rational-distance validator checks the full target witness. Failure of this incomplete search is unknown, never a negative answer; the complete generic oracle handles remaining targets. Its cover kernel was checked against exhaustive optima and below-optimum budgets on 209 small atlas graphs. Earlier generic separator solving independently established an optimum on a 1,596-point target; larger unsuccessful runs are retained as execution failures.

Additional current checks cover twelve side-chain alternatives, all 46 port patterns, and 174 complete fallback layouts. Cross-process tests explicitly exercise the real fallback by exhausting the existing compact allowance at zero: the emitted 330,596-point target has a separately validated witness of size 165,357 and valid recovery under different hash seeds. This source has no clauses. An independent reviewer separately solved a new clause-bearing 12,418-point default target and checked recovery under two further seeds. A 1,292-core-vertex asymmetric source was checked through its accepted default fallback drawing under four seeds, with identical repaired coordinates. Its full target was not expanded because of the observed memory cost. These scopes distinguish branch checks, complete solved targets and proof.

Prerequisites used on 24 September 2026 were uv 0.12.17, Python 3.12.14, NetworkX 3.7, Z3 bindings 5.1.0.0, tsmpy 0.9.3, and Typst 0.15.1. Python dependencies are locked in `uv.lock`; the library versions

are part of the ordering audit. No Lean certificate was requested or produced. Mathlib was not available on the current Lean search path. No remote repository or board publication is part of this result.

From the repository root, use the following commands. Paths below use a shell variable only to keep lines readable.

```
uv sync --locked
campaign=campaigns/planar-three-sat-geometric-connected-domination
uv run --locked python "$campaign/work/check.py" --self-test
uv run --locked python "$campaign/work/check.py" \
  --candidate "$campaign/work/algorithm.py"
uv run --locked python "$campaign/work/verify.py" \
  --candidate "$campaign/work/algorithm.py"
uv run --locked python "$campaign/rounds/018/alternate_witnesses.py"
uv run --locked python "$campaign/rounds/032/test_templates.py"
uv run --locked python "$campaign/rounds/033/test_layout.py"
uv run --locked python "$campaign/rounds/039/check_reconstruction.py"
uv run --locked python "$campaign/rounds/041/certify_negative.py"
uv run --locked python \
  "$campaign/reviews/deterministic-repair/check_recovery.py"
```

The two public map modes consume and emit JSON:

```
uv run --locked python "$campaign/work/algorithm.py" \
  < source.json > target.json
uv run --locked python "$campaign/work/algorithm.py" --extract \
  < recovery-input.json > source-output.json
```

The recovery input has keys `source` and `target_solution`; the latter is a list of distinct point indices or the string `NO-SOLUTION`. Source JSON has keys `variables`, `clauses`, and `embedding`; target JSON has `points` and `K`. The full schemas are in `work/contract.md`. The finite routing certificates are in `rounds/032/templates.json`. Detailed logs are retained under `rounds/039/` and the two review directories, including the original failing hash-order example. Reproducible bulk point dumps are excluded; their source generators, seeds, candidate revisions and output digests remain available.

To rebuild this document and its checked vector figures:

```
uv run --locked python "$campaign/work/figures/draw.py"
typst compile "$campaign/work/manuscript.typ" \
  "$campaign/work/manuscript.pdf"
```